Nicholas Garza

github.com/nickgarza | nicholaspgarza@gmail.com | linkedin.com/in/nicholaspgarza

Experience

Microsoft — Redmond, WA

Software Engineer II - Xbox Graphics — August 2020 - Present

- Automated the next generation Xbox infrastructure to host handheld and pre-release devices for driver build deployment and automated game benchmarking using Azure DevOps and Postgres SQL
- Implemented the validation and debug tooling in C++ for DirectX APIs in the Xbox Game Development Kit
- Enabled high frame rate support and HDR plane composition for the Xbox consoles in the kernel mode driver
- Instrumented user mode driver to capture GPU command buffers and resubmit work to maintain execution timing within nanoseconds across any driver changes in the driver and shader compiler
- Doubled mesh shader throughput by unlocking the thread count limit previously limited to 32-bit integer maximum
- Owned the GPU Pipeline View and implemented debug visualizations for the GPU Performance Investigator for Xbox
- Mentored team interns and intern cohort career development workshops

Medtronic — Northridge, CA

Software Engineer Intern — May 2019 - August 2019

- Developed automated testing framework in Java for the Minimed insulin pump and integrated the framework with the company wide testing platform hosted in Microsoft Azure. Expanded the framework to showcase real time unit testing results and metadata visualization for Android/iOS mobile app crash reporting.

Johns Hopkins Computer Science Department — Baltimore, MD

Head Teaching Assistant — January 2019 - May 2020

- Implemented autograder support for grading student assignments in Python
- Lead team of course assistants to set learning goals for introductory programming course

University of Virginia — Charlottesville, VA

Systems Bioengineering Research Intern — May 2018 - August 2018

- Analyzed terabytes of human genomic data in R to correlate inflammatory cell response with coronary artery diseases
- Developed R package and C++ workflows for filtering and sorting gene expression types
- Accepted to present at 2018 BMES conference in Atlanta, GA and 2018 REUS conference in Alexandria, VA

Technical Skills

Programming Languages Python, C/C++, Java, C#, R

Tools/Frameworks Git, DirectX, *nix, SQL, Windbg, Matlab, JUnit

Education

Johns Hopkins University — Baltimore, MD

B.S. BIOMEDICAL ENGINEERING, B.S. COMPUTER SCIENCE — August 2016 - May 2020

- Honors: CS Honors Graduate, Dean’s List, UPE Honors Society, Hopkins Academic Success Award